Nixor O Levels
Test Week 1
Physics
Grade 10
1001/02

Student Name. _____

Student ID: _____

Section : _____

Time: 60 minutes

Max marks: 30

Please read and follow the instructions below:

- Please write your name, section and ID on the question paper.
- Do not turn over the paper unless instructed to do so.
- Only write in blue or black ink.
- Use of correction fluid is strictly prohibited.
- If you have any questions, raise your hand to ask the invigilator.
- Omission of working will result in loss of marks.
- Use a pencil for making straight lines.
- Answer all questions.
- Calculators are allowed.

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Section A

Answer **all** the questions in this section. **Circle** the correct option with a pencil.

1 The diameter and the length of a thin wire, approximately 1 m in length, are measured as accurately as possible.

What are the best instruments to use?

	diameter	length
A	micrometer	rule
B	micrometer	vernier calipers
C	rule	tape
D	vernier calipers	rule

2 A quantity is quoted as having a value of 6.2 ms.

In what units is it measured?

A metres

B metres per second

C microseconds

D milliseconds

3 A student collects stones and finds their density.

Which apparatus is needed to measure the mass and the volume of the stones?

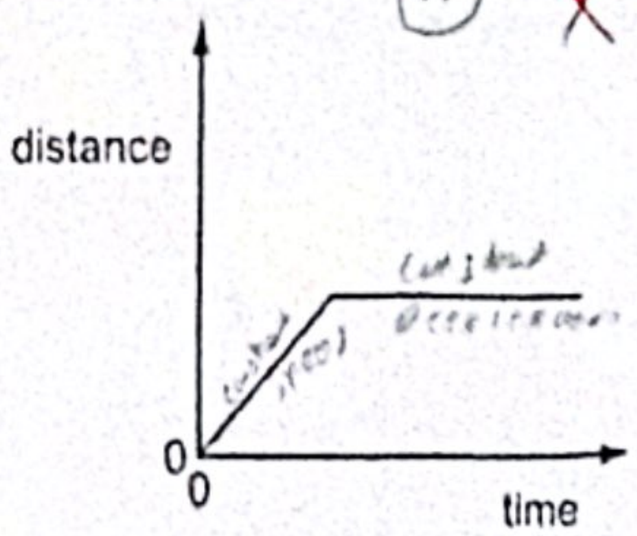
	mass	volume
A	newton meter	measuring cylinder and water
B	newton meter	ruler and calipers
C	top-pan balance	measuring cylinder and water
D	top-pan balance	ruler and calipers

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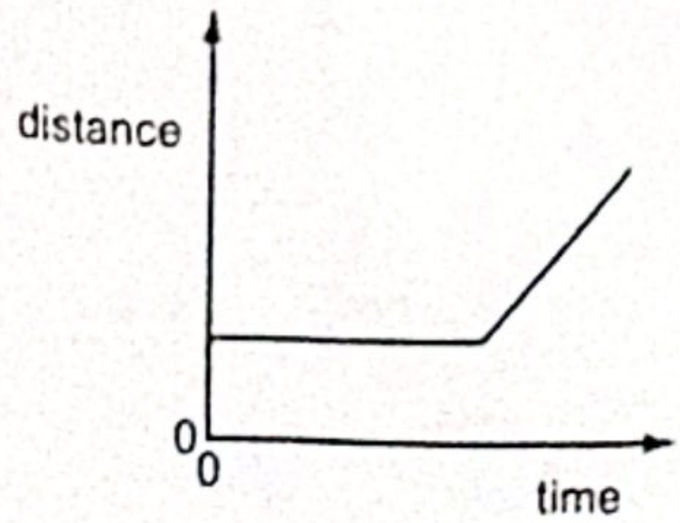
4 An object moves initially with constant speed and then with constant acceleration.

Which graph shows this motion?

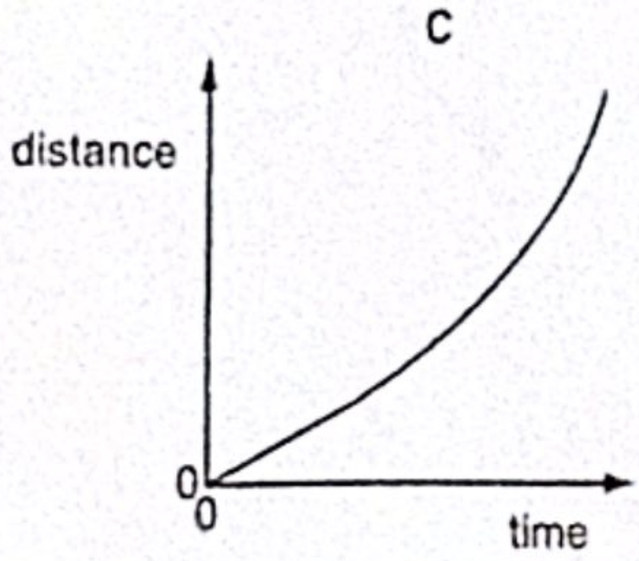
A



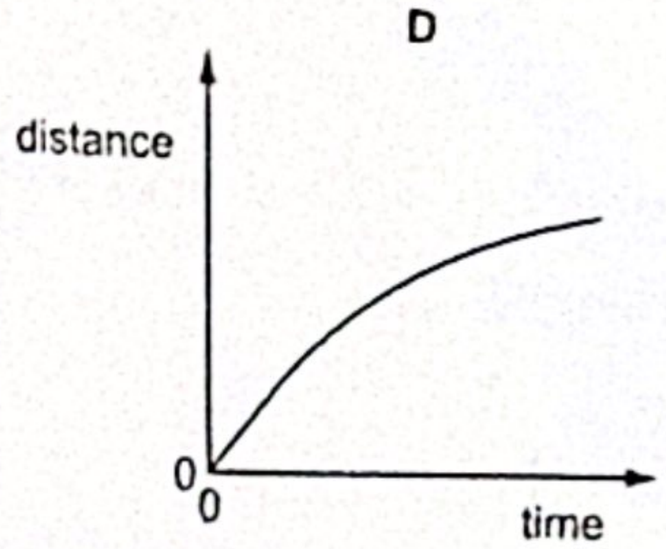
B



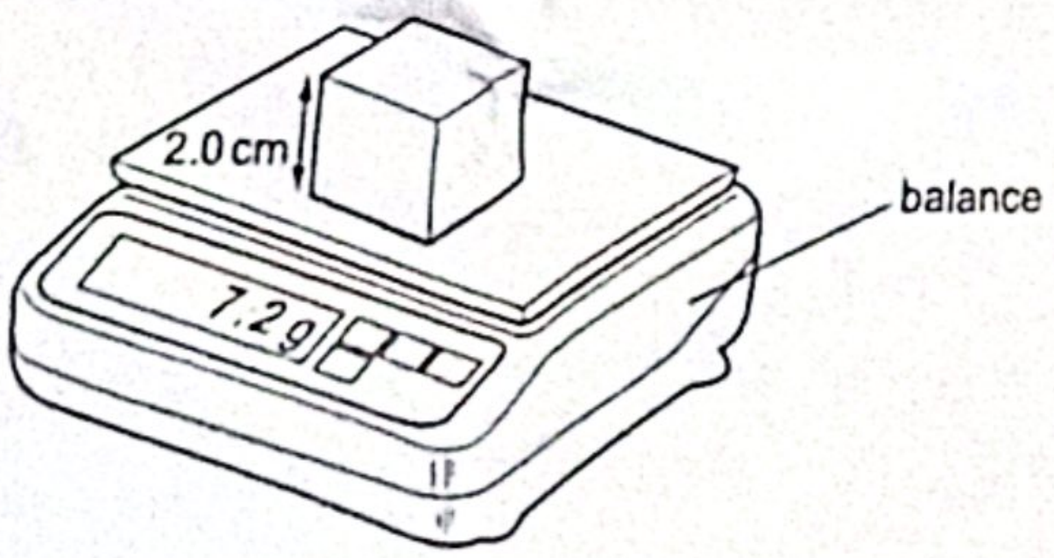
C



D



5 A cube of side 2.0 cm is placed on a balance.



What is the density of the cube?

A 0.90 g/cm³ B 1.2 g/cm³ C 1.8 g/cm³ D 3.6 g/cm³

Section B

Answer **all** the questions in this section. Answer in the spaces provided.

1 Fig. 1.1 shows the path of a football as it is kicked along the ground between three players. The distances between the players are shown on Fig. 1.1.

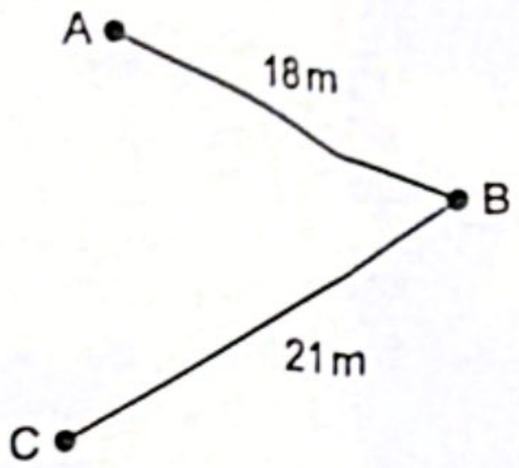


Fig. 1.1

The ball takes 1.2 s to travel from player A to player B.

(i) Calculate the average speed of the ball between A and B.

average speed = distance travelled / time taken.

$V = \frac{\Delta x}{\Delta t}$

$= \frac{18}{1.2}$

$\frac{2}{2}$

average speed = 15 m/s [2]

(ii) Player B kicks the ball to player C.
It travels with the same average speed.
Calculate the time taken for the ball to travel from B to C.

$15 = \frac{21}{t}$ $\quad 15\,m/s = \frac{21\,m}{t}$

$t = \frac{21}{15}$ $\quad t = \frac{21\,m}{15\,m/s}$

$t = 1.4\,s$

$\frac{2}{2}$

time = 1.4 s [2]

(iii) Suggest why the speed of the ball might change during its motion from A to B.

The speed might change due to air resistance which will slow it down the object down and other forces acting upon it. [1]

$\frac{1}{1}$

(iv) Discuss whether the average velocities, from A to B and from B to C, are the same.

The average velocities are not the same because velocity has a direction and here direction is changing. [1]

$\frac{1}{1}$

[Total: 6]

V = speed with direction

	A-B	B-C
mag:	15	15
dir:	east southwest	southeast

PCC

2 The racing car shown in Fig. 2.1 uses a Kinetic Energy Recovery System (KERS). This sy
stores within the car some of the kinetic energy lost when the car slows down for a corner.
driver can later release the stored energy when maximum power is required.

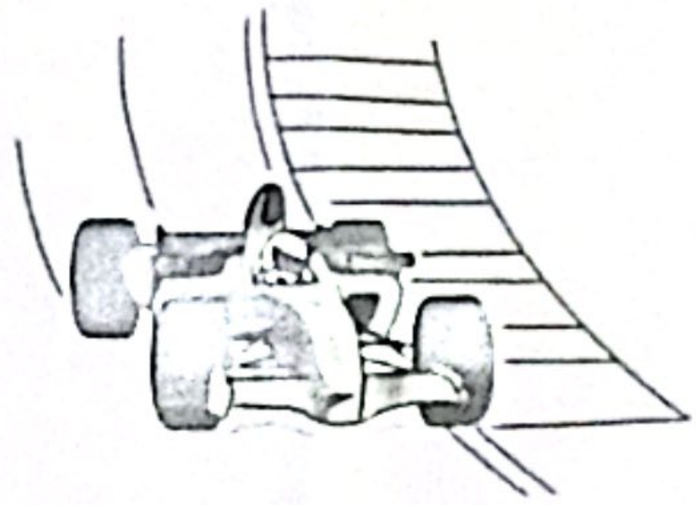


Fig. 2.1

The car approaches a corner and decelerates from 82 m/s to 61 m/s in 0.90 s.
Calculate the deceleration.

$\text{deceleration} = \frac{v-u}{\Delta t} \rightarrow \frac{\text{final - initial velocity}}{\text{change in time}}$

2/2

$a = \frac{61-82}{0.9}$

$a = -23.33$

$\approx -23\ m/s^2$

deceleration = $-23\ m/s^2$ ..... [2]

[Total: 2]

$a = \frac{61-82}{0.9}$

$a = -23\ m/s^2$

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An astronaut has a mass of 65 kg on Earth, where the gravitational field strength is 9.8 N/kg.

(a) Calculate the astronaut's weight on Earth.

weight = mass × gravitational field strength

$W = m \times g$

$W = 65 \times 9.8$

$W = 637\ N$

$\approx 640\ N$

2/2

weight on Earth = ...640 N... [2]

R **(b)** Complete the following sentence.

The astronaut's weight on Earth is the ...gravitational... force between the astronaut and ...the Earth.... [1]

1/1

DCL **(c)** The astronaut undertakes a Moon landing. On the Moon the gravitational field strength is 1.6 N/kg.

(i) State the astronaut's mass on the Moon.

mass = ...65 kg...

DCL **(ii)** Calculate the weight of the astronaut on the Moon.

$W = 65 \times 1.6$

$W = 104$

$\approx 100\ N$

2/2

weight on Moon = ...100 N... [2]

[Total: 5]

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4 Fig. 4.1 is the speed-time graph for a racing car of total mass 650 kg as it sets off from rest at the start of a race.

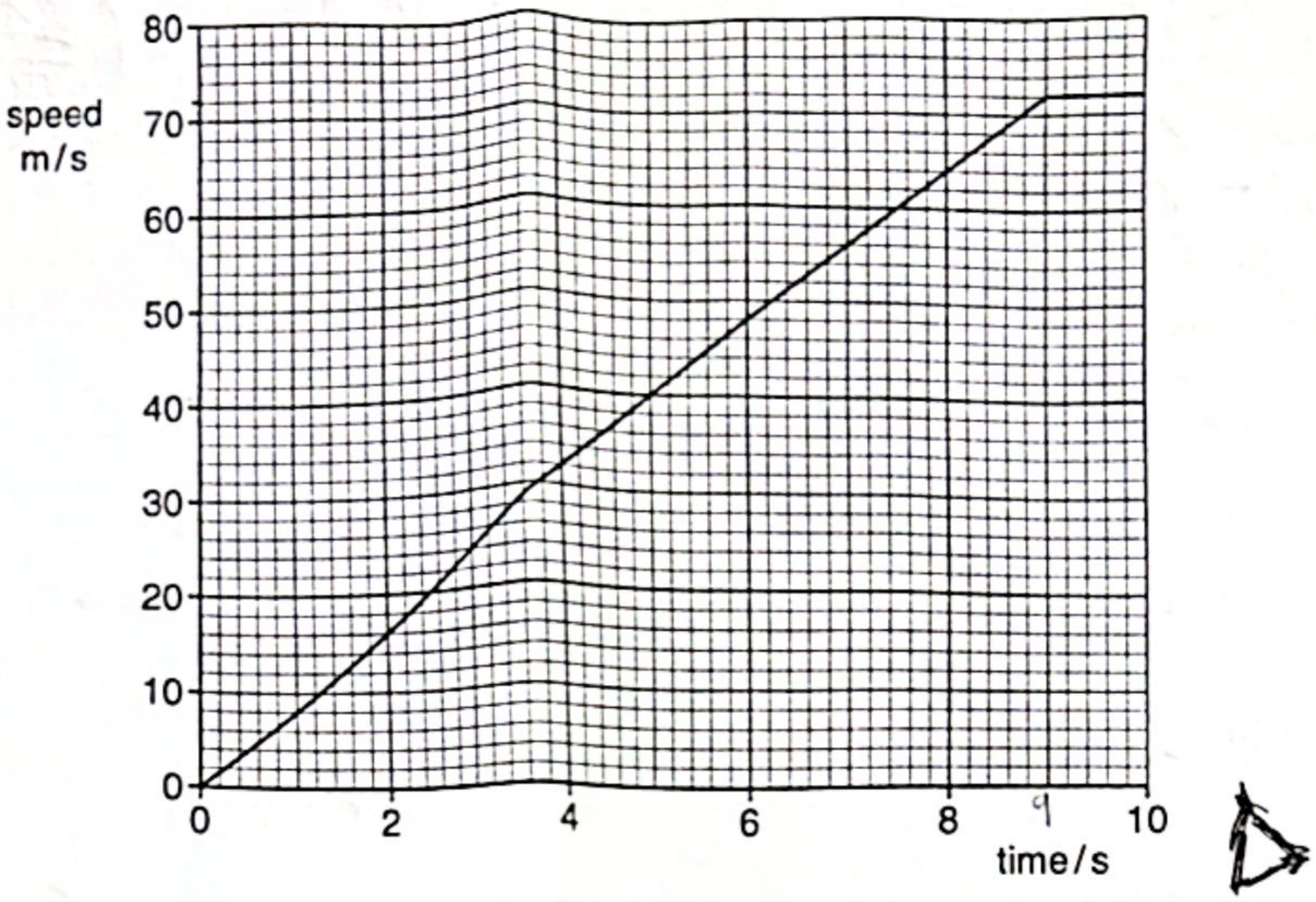


Fig. 4.1

The car travels in a straight line until time $t = 9.0$ s.

(a) State the speed of the car at time $t = 9.0$ s.

speed = 72 m/s .....[1]

(b) Calculate, for the car between times $t = 0$ and $t = 9.0$ s,

(i) the distance travelled, -> area under graph

$\frac{1}{2}(9)(72)$

$= 324$ m

≈ 320 m

distance = 320 m .....[2]

(ii) the acceleration of the car,

acceleration = change in velocity / change in time

$a = \frac{\Delta v}{\Delta t}$

$a = \frac{72 - 0}{9} = 8$ m/s²

acceleration = 8 m/s² .....[2]

(d) After $t = 9.0$ s, the car starts to turn a corner and follows a circular path at constant speed.

(i) Explain why the car is accelerating even though its speed is constant.

The car is accelerating because uniform circular motion requires objects to accelerate to maintain constant speed because direction changing requires more energy to maintain the same speed and due to other forces such as friction and [2] increasing.

[Total: 7]

A student wishes to find the density of a stone. He uses a measuring cylinder and a spring balance with a scale marked in newtons. The measuring cylinder, spring balance and stone are shown in Fig. 5.1.

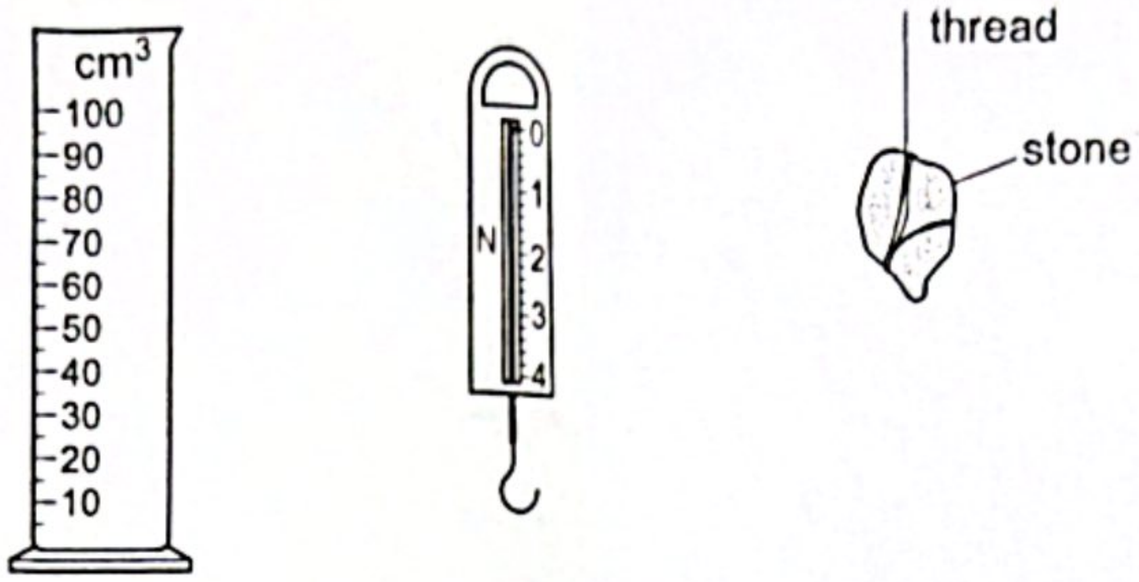


Fig. 5.1

The student knows that the gravitational field strength is 9.8 N/kg.

(a) Describe how the student uses the spring balance to find the mass of the stone.

He makes a ~~hook~~ loop with the thread and puts it on the hook of the spring balance then measures the readings from eye level perpendicular to it. He then converts the weight into mass by using weight = mass × gravity formula and dividing the measured value by the gravitational field strength. [2]

(b) Describe how the student uses the measuring cylinder to find the volume of the stone.

He fills up the ~~volume~~ measuring cylinder with sufficient amount of liquid and then measures its volume (V1). Then he puts the stone ~~he puts the stone in at measures the~~ in the measuring cylinder and records the second volume (V2) before subtracting ~~the second~~ V2-V1 to find the change in volume which is the answer. measure ~~from lowest~~ value from lower meniscus [2]

(c) The mass of the stone is 150 g and its volume is 70 cm^3. Calculate the density of the stone.

$\rho = \frac{M}{V}$

density = $\frac{\text{mass}}{\text{volume}}$

density = $\frac{150\,g}{70\,cm^3}$

density = 2.142 g

$\approx 2.14\ g/cm^3$

$\approx 2.1\ g/cm^3$

density of stone = 2.1 g/cm^3 [1]

[Total: 5]

END OF TEST